

- Conducted research on cloud computing and middleware
- Taught courses *Advanced Internet Computing* and *Distributed Systems Technologies* on graduate level
- Supervised a large number of master students in their thesis work

Research and Teaching Assistant

Oct. 2007 - Oct. 2011

Vienna University of Technology

- Conducted research on service-based systems engineering
- Taught various courses as TA

Research Software Engineer

Jul. 2005 - Sep. 2007

Siemens PSE, Vienna

- Worked on industrial research project in the area of telecommunications and network management

ACQUIRED GRANTS

I have successfully acquired grants (personal as well as project grants) totalling **around 2.5 million EUR**:

- WASP Academic PhD Student (2022-2026)
Academic Supervisor
Grant value $\approx 3,600,000$ SEK
(≈ 355.000 EUR)
- Debloating AI Systems (2021), AoA SEED Grant
Co-applicant
Total budget 296,000 SEK (my part $\approx 128,000$ SEK)
(≈ 13.000 EUR)
- FFI TrafCloud (2019 – 2021), VINNOVA
Co-applicant
Chalmers budget 5,500,000 SEK (my part $\approx 3,200,000$ SEK)
(≈ 315.000 EUR)
- VR Starting Grant (2019 – 2022), VR
Sole applicant
Grant value 3,639,000 SEK
(≈ 359.000 EUR)
- WASP Academic PhD Student (2017 – 2021), WASP
Academic Supervisor
Grant value $\approx 3,600,000$ SEK
(≈ 355.000 EUR)
- Chalmers AoA Hiring Startup Package (2017 – 2021), Chalmers Foundation
Sole recipient
Grant value 8,000,000 SEK
(≈ 789.000 EUR)

- Individual Project “MINCA” (2016 – 2019), Swiss National Science Foundation
Sole applicant
Grant value 180,000 CHF, at University of Zurich
($\approx$ 163.000 EUR)
- Technology Transfer Project “DevCloud” (2016 – 2019), Hasler Foundation
Co-applicant
Grant value 50,000 CHF, at University of Zurich
($\approx$ 45.000 EUR)

RESEARCH VISITS

Institute of of High-Performance Computing, Singapore
University of Stuttgart, Germany

Feb./Mar. 2013
Apr. 2010

INVITED TALKS AND SEMINARS (SELECTED LIST)

- *Bots in Software Engineering*, Invited Talk at WASP4All Webinar, May 2022
- *Unit testing performance using code microbenchmarks. How far are we?*, Invited Talk at the MongoDB Inc. Performance Seminar, November 2020
- *Performance stability of (public) clouds*, Invited Talk at Vehicle Electronics and Connected Services (VECS, industrial conference), April 2019
- *AWS Lambda and #serverless. What’s all the fuzz about?*, Invited Talk at the Chaos Engineering Days Stockholm, December 2018
- *Software performance testing in a public cloud – how bad is it really?*, Invited Talk at the Metrics Day 2018, October 2018
- *Performance Testing of and in the Cloud*, Invited Lecture at the IDEA League Doctoral School on Engineering Complex Systems and IT with Big data and Information Technology, September 2018
- *Predictability of Performance in Public Clouds - Some Empirical Data and Lessons Learned for Software Performance Testing*, Seminar at IBM Research, Ireland, November 2017
- *Predictability of Performance in Public Clouds - Some Empirical Data and Lessons Learned for Software Performance Testing*, Invited Talk at the Sixth International Workshop on Load Testing and Benchmarking of Software Systems (LTB 2017), co-located with ICPE’17, April 2017
- *Your User is the Canary - Practices of and Obstacles to Conducting Experiments in Production*, CHOOSE Forum, University of Zurich, September 2016
- *How Predictable is the Performance of IaaS Instances? Some Insights from Benchmarking EC2, GCE, Azure, and Softlayer*, Open Cloud Day, ZHAW Winterthur, June 2016
- *Software Development for the Cloud – Challenges and Opportunities*, Faculty of Informatics, University of Lugano, February 2016

- *Software Development for the Cloud – Challenges and Opportunities*, VU University Amsterdam, November 2015
- *Trends, Opportunities and Challenges of Software Development for the Cloud*, Indian Institute of Technology Ropar, November 2015
- *Trends, Opportunities and Challenges of Software Development for the Cloud*, Indian Institute of Technology New Delhi, November 2015
- *Software Development for the Cloud – Challenges and Opportunities*, Chalmers, Gothenburg, June 2015
- *Engineering Java Applications for the Infrastructure-as-a-Service Cloud*, software evolution & architecture lab, University of Zurich, Zurich, October 2013
- *Building Applications for the Infrastructure-as-a-Service Cloud with CloudScale*, IMDEA Software Institute, Madrid, April 2013
- *Building Elastic Java Applications based on CloudScale and the Infrastructure-as-a-Service Paradigm*, National University of Singapore, School of Computing, March 2013
- *Building Elastic Java Applications based on CloudScale and the Infrastructure-as-a-Service Paradigm*, University of Illinois Advanced Digital Sciences Center, Singapore, February 2013

AWARDS

- Distinguished Reviewer Award (ICSE 2023)
- Best Reviewer Award (SANER 2022)
- Distinguished Reviewer Award (ICSME 2020)
- Distinguished Reviewer Award (MSR 2020)
- Outstanding Reviewer Award (ICPE 2020)
- Outstanding Reviewer Award (ICPE 2019)
- Best Student Paper Award (Middleware 2016)
- Best Short Paper Award (ICPC 2016)

DEPARTMENT SERVICE

- Unit Manager Software Engineering 2 (Chalmers and the University of Gothenburg, 01.04.2022 - ongoing)
- Master Program Administrator Software Engineering (Chalmers and the University of Gothenburg, 01.03.2021 - 01.08.2021)
- Member of the Faculty Senate Drafting Board (IT Faculty, University of Gothenburg, 2021 - 2024)

- Postdoc Representative at the Department of Informatics, University of Zurich (2015 - 2016)
- Member of the Habilitation Committee of Ivona Brandic (TU Vienna, 2013)
- Member of the Habilitation Committee of Karl Michael Göschka (TU Vienna, 2012)

PROFESSIONAL SERVICE

Organized Events

- Co-Organizer of the 1st and 2nd *Vienna Software Seminar (VSS)* on the Relation of Software Architecture and DevOps/Continuous Delivery. Organizers: Uwe Zdun, Philipp Leitner, Florian Rosenberg, Erik Wittern.
- Member of the Steering Committee of the International Workshop on Quality-Aware DevOps (QUDOS). From 2018.
- Dagstuhl GI Seminar on *Software Performance Engineering in the DevOps World* (September 26th – September 30th 2016, Schloss Dagstuhl, Germany). Co-organized with Andre van Hoorn, Pooyan Jamshidi, and Ingo Weber.
- Half-day *CloudWave Training Event* (September 5th 2016, TU Vienna, Austria). Half-day session at ESOC'16. Co-organized with Andreas Metzger and Eliot Salant.

Editorial Memberships

- Associate Editor of Springer Empirical Software Engineering (EMSE)
- Academic Editor of PeerJ Computer Science
- Editorial Board Member of Elsevier's Journal on Systems and Software (JSS). Responsible for coordinating and implementing the publicity efforts of the journal. 2017 - 2019.
- Special Issue Editor for "Converging Fog and Cloud Computing", in IEEE Cloud Computing. Co-edited with Erik Elmroth, Stefan Schulte, and Srikumar Venugopal.
- Special Issue Editor for "Software Tools and Technologies for Monitoring and Prediction of Cloud Services", in Software: Practice and Experience (published by Wiley-Blackwell). Co-edited with Rajiv Ranjan, Rajkumar Buyya, Armin Haller, and Stefan Tai

Chairmanships

- Chair of the Selection Committee of the 2022 SPEC Kaivalya Dixit Distinguished Dissertation Award
- Program Co-Chair of 13th ACM/SPEC International Conference on Performance Engineering (ICPE)
- Awards Co-Chair of 12th ACM/SPEC International Conference on Performance Engineering (ICPE)
- Publication Chair of 10th ACM/SPEC International Conference on Performance Engineering (ICPE)

- PC Co-Chair of the 3rd International Workshop on Quality-Aware DevOps (QUDOS), co-located with the 8th ACM/SPEC International Conference on Performance Engineering (ICPE 2017)
- Co-Chair of a Special Session on Services Computing and Internet Technologies (SerCo 2016) at the International Conference on High Performance Computing & Simulation (HPCS 2016)
- Tutorial Chair of the 16th International Conference on Web Engineering (ICWE2016)
- Workshop Chair of the European Conference on Service-Oriented and Cloud Computing (ESOCC) 2015.
- General Chair of the 1st International Workshop on Monitoring and Prediction of Cloud Services (MoPoC'12), co-located with ICSOC'12.

Reviews for Funding Agencies

(partial list)

- Dutch Research Council (NWO)
- Natural Sciences and Engineering Research Council of Canada (NSERC)
- Deutscher Akademischer Austauschdienst (DAAD)
- Luxembourg National Research Fund (FNR)
- Wallenberg AI, Autonomous Systems, and Software Program (WASP)

Program Committee Memberships

(partial list)

- International Conference on Fundamentals of Software Engineering (FSE 2024)
- International Conference on Software Engineering (ICSE 2023)
- IEEE/ACM International Conference on Automated Software Engineering (ASE 2022 - 2023)
- IEEE International Conference on Cloud Engineering (IC2E 2021)
- European Symposium on Serverless Computing and Applications (ESSCA 2020)
- International Conference on Microservices (MICROSERVICES 2020)
- IEEE/ACM International Conference on Automated Software Engineering - Demonstration Track (ASE 2019)
- IEEE International Conference on Software Analysis, Evolution and Reengineering (SANER, from 2019)
- IEEE International Conference on Software Maintenance and Evolution (ICSME, from 2018)
- International Conference on Mining Software Repositories (MSR, from 2018)

- ACM International Conference on Distributed Event-Based Systems (DEBS 2018)
- ACM/SPEC International Conference on Performance Engineering (ICPE 2018, 2020)
- IEEE International Conference on Cloud Computing (CLOUD, 2015 – 2018)
- International Conference on the Quality of Information and Communications Technology (QUATIC, 2018)
- International Conference on Edge Computing (EDGE 2017)
- International Symposium on Fog and Edge Computing (ISFEC 2017)
- Euromicro Conference on Software Engineering and Advanced Applications (SEAA, 2018)
- International Conference on Service-Oriented Computing (ICSOC, 2015 – 2017)
- European Conference on Service-Oriented and Cloud Computing (ESOCC, 2015, 2017)
- International Conference on Web Engineering (ICWE, 2015, 2019)
- IEEE/ACM International Utility and Cloud Computing Conference (UCC, 2014 – 2017)
- International Conference on Service Oriented Computing and Applications (SOCA, 2014 – ongoing)
- Cloud Challenge, held in conjunction with UCC (2014)
- International Workshop on RESTful Designs (WS-REST, co-located with WWW, 2018)
- International Workshop on Software Engineering Aspects of Continuous Development and New Paradigms of Software Production and Deployment (DEVOPS, 2018)
- International Workshop on Autonomous Control for Performance and Reliability Trade-offs in Internet of Services (ACPROSS, 2017)
- International Workshop on Inter-Organizational Processes (IOPs, 2016)
- International Workshop on Big Data Software Engineering (BIGDSE, 2015 – ongoing)
- EAI International Conference on Cloud, Networking for IoT Systems (CN4IoT, 2015)
- International Conference on Internet and Web Applications and Services (ICIW, 2014 – 2015)
- International Conference on the Future Internet of Things and Cloud (FiCloud, 2014)
- International Conference on Service Oriented Computing (ICSOC) Demonstration Track (2014 – ongoing)
- International Workshop on Federative and Interoperable Cloud Infrastructures (FedICI, 2014)
- International Workshop on Engineering Cloud Applications & Services (EnCASE, 2014)
- Central-European Workshop on Services and their Composition (ZEUS, 2012 – ongoing)
- International Workshop on Evolutionary Business Processes EVL-BP), co-located with EDOC (2011 – 2013)

- International Workshop on Principles of Engineering Service-Oriented Systems (PESOS), co-located with ICSE (2013)
- Track on Provisioning and Management of Service Oriented Architecture and Cloud Computing (PROMASC), a Track of the WETICE (2011 – 2012)
- International Workshop on Performance Assessment and Auditing in Service Computing (PAASC), co-located with ICSOC (2011 – 2012)
- Workshop on Emerging Web Services Technology (WEWST), co-located with ECOWS (2009 – 2011)
- International Workshop on Dynamic and Declarative Business Processes (DDBP), co-located with EDOC (2010)

Reviews for Journals

(partial list)

- Elsevier Information and Software Technology (IST)
- IEEE Transactions on Software Engineering (TSE)
- Journal of Systems and Software (JSS)
- Springer Empirical Software Engineering (ESEM)
- ACM Transactions on Internet Technology (TOIT)
- IEEE Transactions on Cloud Computing (TCC)
- Wiley Concurrency and Computation: Practice and Experience
- Oxford Press: The Computer Journal
- International Journal of Communication Networks and Distributed Systems (IJCNDS)
- Elsevier Information Processing Letters (IPL)
- IEEE Internet Computing (IC)
- IEEE Software
- Elsevier Journal of Parallel and Distributed Computing (JPDC)
- ACM Transactions on the Web (TWEB)
- Elsevier Journal of Systems and Software (JSS)
- Springer Computing (Archives for Scientific Computing)
- Journal of Computer Science and Technology (JCST)
- International Journal of Cooperative Information Systems (IJCIS)
- IEEE Transactions on Services Computing (TSC)
- Transactions on Pattern Languages of Programming (TPLoP)

RESEARCH STUDENTS

Current PhD Students

- **Ranim Khojah** (Large Language Models in Software Engineering)
- **Huaifeng Zhang** (Debloating ML Systems, *co-supervisor*)
- **Hazem (Peter) Samoaa** (ML for Performance Prediction)
- **Hamdy Michael Ayas** (Architectural Decision Making in Cloud Computing)
- **Linda Erlenhov** (Bots in Software Development)

Graduated PhD Students

- **Joel Scheuner** (Cloud Benchmarking, now software engineer at LocalStack, Switzerland)
- **Christoph Laaber** (UZH, microbenchmarking in CI builds, co-supervised with Harald C. Gall, now postdoc at Simula, Norway))
- **Gerald Schermann** (UHZ, continuous experimentation, co-supervised with Harald C. Gall, now release engineer at digitec.ch, Switzerland)
- **Jürgen Cito** (UZH, cloud performance engineering, co-supervised with Harald C. Gall, now Assistant Professor at TU Wien, Austria)

PhD Committee Appointments

- **César Soto-Valero** (KTH, member of grading committee)
- **Mohammad Reza Saleh Sedghpour** (Umeå University, member of grading committee)
- **Alim Ul Gias** (Imperial College London, external examiner)
- **Luca Traini** (Università degli Studi dell'Aquila, assessor)
- **Hugo Sica de Andrade** (Chalmers University of Technology, substitute member of grading committee)
- **Lars Larsson** (Umeå University, faculty opponent)
- **Alfred Åkesson** (Lund University, licenciate discussion leader)

FORMAL PEDAGOGICAL TRAINING

- **Diversity and inclusion in higher education** (2018, 2 ECTS)
- **University teaching and learning** (2017, 3 ECTS)
- **Enhancing learning through writing** (2018, 5 ECTS)
- **Perspectives on learning** (2018, 2.5 ECTS)
- **Supervision of research** (2017, 3 ECTS)

TEACHING ACTIVITIES

Teaching at Chalmers | GU

- Distributed Systems Mini-Project (DIT356) (since 2023, $\approx$ 80 students)
- Architectures of Scale-Out Systems (DAT490) (since 2022, $\approx$ 40 students)
- Mobile and Web Development (DIT341) (since 2019, $\approx$ 100 students)
- Data Management (DIT032/DAT335) (2018 – 2021, $\approx$ 150 students)

Previous Teaching at University of Zurich

- Informatik 1 (fall 2014 – 2016, $\approx$ 250 students)
- Advanced Software Engineering (spring 2014 – 2017, 10 – 20 students)
- Seminar Advanced Software Engineering (fall 2015 – 2016, 10 – 20 students)
- Master Basismodul (2014 – 2017)
- Informatik-Vertiefung (2014 – 2017)

Previous Teaching at TU Vienna

- VU 2.0 Advanced Internet Computing (fall 2008 - 2013, 140 students)
- VU 4.0 Distributed Systems Technologies (spring 2009 - 2013, 140 students)
- PR 4.0 Internet Computing Lab Work (spring 2009 - 2013)
- UE 2.0 Verteilte Systeme (fall 2009 - fall 2010, $\approx$ 400 students)
- VU 2.0 Grid Computing (spring 2008 - spring 2009, 10 students)

Thesis Supervision at Chalmers | University of Gothenburg

(partial list)

- Florian Johan Markusse: The impact of benchmarking bots on Open-Source-Software projects (master's thesis, co-supervisor, TU Eindhoven, 2021)
- Niclas Alexandersson: JUnit-to-JMH - Automated Generation of Performance Benchmarks from Existing Unit Tests in Java (master's thesis, 2021)
- Sebastian Hafström and Naren Hari Hara Krishnan: Using Policy Engines to handle Authorization in Microservice Architectures (master's thesis, 2021)
- Erik Bäcknas and Eric Börjesson: Total Cost of Ownership Framework to Estimate the Cost of Cloud Computing Services (master's thesis, 2021)
- Yan Wang and Mihkel Sildnik: Debloating Machine Learning Systems (master's thesis, co-advisor, 2021)

- Erik Tingström and George Karageozian: Mining GitHub to Evaluate the Impact of Social Bots on New Committer Retention in OSS Projects (bachelor thesis, 2021)
- Elsada Lagumdžić: Evaluation of Open Source Function as a Service Frameworks: An Experimental Analysis of Auto Scaling Characteristics (bachelor thesis, 2020)
- Joel Sanderöd Roxell and Matthias Lundell: Automatic container profiling and resource prediction using container orchestration (master's thesis, 2019)
- Yunfang Guo: The Impact of Adopting Continuous Integration on the Delivery Time of Pull Requests - A Partial Replication and Extension (master's thesis, 2019)
- Joel Rollny: An Extensible Framework for Large-Scale Mining of Performance Microbenchmarks (master's thesis, 2018)
- Viktor Holmqvist and Jonathan Nilsfors: Cachematic - Automatic Invalidation in Application-Level Caching Systems (master's thesis, 2018)

ACADEMIC JOURNAL PAPERS [scholar]

- H. M. Ayas, **P. Leitner**, and R. Hebig: *An Empirical Study of the Systemic and Technical Migration Towards Microservices*, Empirical Software Engineering (EMSE), 2023. To appear.
- M. Grambow, D. Kovalev, C. Laaber, **P. Leitner**, and D. Bermbach: *Using Microbenchmark Suites to Detect Application Performance Changes*, IEEE Transactions on Cloud Computing (TCC), 2022. To appear.
- M. Jangali, Y. Tang, N. Alexandersson, **P. Leitner**, J. Yang, and W. Shang: *Automated Generation and Evaluation of JMH Microbenchmark Suites from Unit Tests*, IEEE Transactions on Software Engineering (TSE), 2022. To appear.
- F. Markusse, **P. Leitner**, and A. Serebrenik: *Using Benchmarking Bots for Continuous Performance Assessment*, IEEE Software, 2022. To appear.
- H. Samoaa, F. Bayram, P. Salza, and **P. Leitner**: *A Systematic Mapping Study of Source Code Representation for Deep Learning in Software Engineering*, IET Software, 2022.
- L. Erlenhov, F. Gomes, and **P. Leitner**: *Dependency management bots in open-source systems—prevalence and adoption*. PeerJ Computer Science 8:e849 (2022). <https://doi.org/10.7717/peerj-cs.849>
- C. Laaber, **P. Leitner**, and H. Gall: *Applying Test Case Prioritization to Software Microbenchmarks*, Empirical Software Engineering, 26, 133 (2021). <https://doi.org/10.1007/s10664-021-10037-x>
- M. Grambow, C. Laaber, **P. Leitner**, and D. Bermbach: *Using Application Benchmark Call Graphs to Quantify and Improve the Practical Relevance of Microbenchmark Suites*, PeerJ Computer Science 7:e548 (2021). <https://doi.org/10.7717/peerj-cs.548>
- J. Scheuner and **P. Leitner**: *Function-as-a-Service Performance Evaluation: A Multivocal Literature Review*, Journal of Systems and Software (JSS), vol. 170. 2020.
- Y. Guo and **P. Leitner**: *Studying the Impact of CI on Pull Request Delivery Time in Open Source Projects — a Conceptual Replication*, PeerJ Computer Science, 2019.
- D. Costa, CP. Bezemer, **P. Leitner**, and A. Andrzejak: *What’s Wrong With My Benchmark Results? Studying Bad Practices in JMH Benchmarks*, IEEE Transactions on Software Engineering (TSE). Early access, doi: 10.1109/TSE.2019.2925345.
- B. Varghese, **P. Leitner**, S. Ray, K. Chard, A. Barker, Y. Elkhatib, H. Herry, C.H. Hong, J. Singer, F.P. Tso, E. Yoneki, and M.F. Zhani: *Cloud Futurology*, IEEE Computer 52(9): 68-77 (2019)

- C. Laaber, J. Scheuner, and **P. Leitner**: *Software Microbenchmarking in the Cloud. How Bad is it Really?*, Empirical Software Engineering 24(4): 2469-2508 (2019)

- **P. Leitner**, E. Wittern, J. Spillner, and W. Hummer: *A Mixed-Method Empirical Study of Function-as-a-Service Software Development in Industrial Practice*, Journal of Systems and Software 149: 340-359 (2019)
- G. Schermann, J. Cito, **P. Leitner**, U. Zdun, and H. Gall: *We're Doing It Live: A Multi-Method Empirical Study on Continuous Experimentation*, Information and Software Technology 99: 41-57 (2018)
- G. Schermann, J. Cito, and **P. Leitner**: *Continuous Experimentation - Challenges, Implementation Techniques, and Current Research*, IEEE Software 35(2): 26-31 (2018)
- O. Skarlat, M. Nardelli, S. Schulte, M. Borkowski, and **P. Leitner**: *Optimized IoT Service Placement in the Fog*, Service Oriented Computing and Applications 11(4): 427-443 (2017)

- **P. Leitner** and J. Cito: *Patterns in the Chaos – a Study of Performance Variation and Predictability in Public IaaS Clouds*, ACM Transactions on Internet Technology (TOIT), vol. 16, iss. 3, 2016.

- R. Zabolotnyi, **P. Leitner**, W. Hummer, and S. Dustdar: *JCloudScale: Closing the Gap Between IaaS and PaaS*, ACM Transactions on Internet Technology (TOIT), vol. 15, iss. 3, p. 10:1–10:20, 2015.
- J. Cito, D. Gotowka, **P. Leitner**, R. Pelette, D. Suljoti, and S. Dustdar: *Identifying Web Performance Degradations through Synthetic and Real-User Monitoring*, Journal of Web Engineering, 2015.
- A. Metzger, **P. Leitner**, D. Ivanovic, E. Schmieders, R. Franklin, M. Carro, S. Dustdar, and K. Pohl: *Comparing and Combining Predictive Business Process Monitoring Techniques*, IEEE Transactions on Systems, Man, and Cybernetics: Systems, vol. 45, no. 2, pp. 276 - 290. 2014.
- C. Inzinger, W. Hummer, B. Satzger, **P. Leitner**, and S. Dustdar: *Generic Event-Based Monitoring and Adaptation Methodology for Heterogeneous Distributed Systems*, Software: Practice and Experience, 2014.
- **P. Leitner**, W. Hummer and S. Dustdar: *Data-Driven and Automated Prediction of Service Level Agreement Violations in Service Compositions*, Distributed and Parallel Databases (Springer), vol. 31, no. 3, pp. 447 - 470. 2013
- W. Hummer, O. Raz, O. Shehory, **P. Leitner**, and S. Dustdar: *Testing of Data-Centric and Event-Based Dynamic Service Compositions*, Software Testing, Verification and Reliability, vol. 23, iss. 6, pp. 465-497, 2013.
- **P. Leitner**, W. Hummer and S. Dustdar: *Cost-Based Optimization of Service Compositions*, IEEE Transactions on Services Computing (TSC). 2011, doi:10.1109/TSC.2011.53

- W. Hummer, **P. Leitner** and S. Dustdar: *SEPL - A Domain-Specific Language and Execution Environment for Protocols of Stateful Web Services*, Distributed and Parallel Databases (DPD), vol. 29, nr. 4, pp. 277-307. 2011, doi:10.1007/s10619-011-7079-6
- B. Wetzstein, **P. Leitner**, F. Rosenberg, S. Dustdar and F. Leymann: *Identifying Influential Factors of Business Process Performance Using Dependency Analysis*, Enterprise Information Systems (EIS), vol. 5, nr. 1. 2011, doi:10.1080/17517575.2010.493956
- A. Michlmayr, F. Rosenberg, **P. Leitner** and S. Dustdar: *End-to-End Support for QoS-Aware Service Selection, Binding and Mediation in VRESCo*, IEEE Transactions on Services Computing (TSC), vol. 3, nr. 3, pp. 193-205. 2010, doi:10.1109/TSC.2010.20
- A. Michlmayr, F. Rosenberg, **P. Leitner** and S. Dustdar: *Selective Service Provenance in the VRESCo Runtime*, International Journal on Web Services Research (JWSR), vol 7, no. 2, pp. 65-86, 2010, doi: 10.4018/jwsr.2010040104
- **P. Leitner**, F. Rosenberg, S. Dustdar, Daios: *Efficient Dynamic Web Service Invocation*, IEEE Internet Computing, vol. 13, no. 3, pp. 72-80, May/June 2009, doi:10.1109/MIC.2009.57

ACADEMIC CONFERENCE PAPERS [\[scholar\]](#)

- J. Scheuner, R. Deng, J.-P. Steghöfer and **P. Leitner**: *CrossFit: Fine-grained Benchmarking of Serverless Application Performance across Cloud Providers*, in Proceedings of the 15th IEEE/ACM International Conference on Utility and Cloud Computing (UCC). 2022.
- H. Samoa, A. Longa, M. Mohamad, M. H. Chehrehghani, and **P. Leitner**: *TEP-GNN: Accurate Execution Time Prediction of Functional Tests using Graph Neural Networks*, in Proceedings of the 22nd International Conference on Product-Focused Software Process Improvement (PROFES). 2022.
- J. Scheuner, M. Bertilsson, O. Gronqvist, H. Tao, H. Lagergren, JP. Steghöfer and **P. Leitner**: *TriggerBench: A Performance Benchmark for Serverless Function Triggers* (Short Paper), in Proceedings of the International Conference on Cloud Engineering (IC2EE). 2022.
- H. M. Ayas, F. Gomes, and **P. Leitner**: *Empirical Analysis of Microservices Systems Using Consumer-Driven Contract Testing*, in Proceedings of the Euromicro Conference on Software Engineering and Advanced Applications (SEAA). 2022.
- H. M. Ayas, **P. Leitner**, and R. Hebig: *A Grounded Theory study on the migration journey towards microservices*, in Proceedings of the 22nd International Conference on Product-Focused Software Process Improvement (PROFES). 2021.
- H. M. Ayas, R. Hebig, and **P. Leitner**: *Facing the Giant: a Grounded Theory Study of Decision-Making in Microservices Migrations*, in Proceedings of the 15th ACM/IEEE International Symposium on Empirical Software Engineering and Measurement (ESEM). 2021.
- P. Samoa and **P. Leitner**: *An Exploratory Study of the Impact of Parameterization on JMH Measurement Results in Open-Source Projects*, in Proceedings of the 2021 ACM/SPEC International Conference on Performance Engineering (ICPE). 2021.

- C. Laaber, S. Würsten, H. C. Gall, and **P. Leitner**: *Dynamically Reconfiguring Software Microbenchmarks: Reducing Execution Time Without Sacrificing Result Quality*, in Proceedings of the 2020 ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering (ESEC/FSE). 2020.
- L. Erlenhov, F. Gomes, and **P. Leitner**: *An Empirical Study of Bots in Software Development - Characteristics and Challenges from a Practitioner's Perspective*, in Proceedings of the 2020 ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering (ESEC/FSE). 2020.
- G. Schermann, F. Oliveira, E. Wittern, and **P. Leitner**: *Topology-aware Continuous Experimentation in Microservice-based Applications*, in Proceedings of the 2020 International Conference on Service Oriented Computing (ICSOC). 2020.
- J. Cito, **P. Leitner**, M. Rinard, and H. C. Gall: *Interactive Production Performance Feedback in the IDE*, in Proceedings of the 41st ACM/IEEE International Conference on Software Engineering (ICSE). 2019.
- V. Holmqvist, J. Nilsfors, and **P. Leitner**: *Cachematic – Automatic Invalidation in Application-Level Caching Systems*, in Proceedings of the 2019 ACM/SPEC International Conference on Performance Engineering (ICPE) – Industry Track. 2019.
- G. Schermann and **P. Leitner**: *Search-Based Scheduling of Experiments in Continuous Deployment*, in Proceedings of the 34rd IEEE International Conference on Software Maintenance and Evolution (ICSME). 2018.
- C. Laaber and **P. Leitner**: *An Evaluation of Open-Source Software Microbenchmark Suites for Continuous Performance Assessment*, in Proceedings of the 15th International Conference on Mining Software Repositories (MSR), 2018.
- C. Vassallo, G. Schermann, F. Zampetti, D. Romano, **P. Leitner**, A. Zaidman, M. Di Penta, and S. Panichella: *A Tale of CI Build Failures: an Open Source and a Financial Organization Perspective*, in Proceedings of the 33rd IEEE International Conference on Software Maintenance and Evolution (ICSME). 2017
- G. Mazlamo, J. Cito, and **P. Leitner**: *Extraction of Microservices from Monolithic Software Architectures*, in Proceedings of the 24th IEEE International Conference on Web Services (ICWS) - Applications Track. 2017
- J. Cito, G. Schermann, E. Wittern, **P. Leitner**, S. Zumberi, and H. C. Gall: *An Empirical Analysis of the Docker Container Ecosystem on GitHub*, in Proceedings of the 14th International Conference on Mining Software Repositories (MSR), 2017.
- T. Rausch, W. Hummer, **P. Leitner**, and S. Schulte: *An Empirical Analysis of Build Failures in the Continuous Integration Workflows of Java-Based Open-Source Software*, in Proceedings of the 14th International Conference on Mining Software Repositories (MSR), 2017.

- C. Davatz, C. Inzinger, J. Scheuner, and **P. Leitner** : *An Approach and Case Study of Cloud Instance Type Selection for Multi-Tier Web Applications*, in Proceedings of the 17th IEEE/ACM International Symposium on Cluster, Cloud and Grid Computing (CCGrid), 2017.
- J. Cito, F. Oliveira, **P. Leitner**, P. Nagpurkar, and H. C. Gall: *Context-Based Analytics – Establishing Explicit Links between Runtime Traces and Source Code*, In Proceedings of the 38th International Conference on Software Engineering (ICSE'17) – Software Engineering in Practice (SEIP) Track, 2017.
- **P. Leitner** and CP. Bezemer: *An Exploratory Study of the State of Practice of Performance Testing in Java-Based Open Source Projects*, in Proceedings of the 7th ACM/SPEC International Conference on Performance Engineering (ICPE), 2017.

- G. Schermann, D. Schöni, **P. Leitner**, and H. C. Gall: *Bifrost – Supporting Continuous Deployment with Automated Enactment of Multi-Phase Live Testing Strategies*, in Proceedings of the 2016 ACM/IFIP/USENIX Middleware Conference. **Best Student Paper Award**.

- **P. Leitner**, J. Cito, and E. Stöckli: *Modelling and Managing Deployment Costs of Microservice-Based Cloud Applications*, in Proceedings of the 9th IEEE/ACM International Conference on Utility and Cloud Computing (UCC), 2016.
- O. Skarlat, S. Schulte, M. Borkowski, and **P. Leitner**: *Resource Provisioning for IoT Services in the Fog*, in Proceedings of the 9th IEEE International Conference on Service Oriented Computing & Applications (SOCA), 2016.
- **P. Leitner** and J. Scheuner: *Bursting With Possibilities – an Empirical Study of Credit-Based Bursting Cloud Instance Types*, in Proceedings of the 8th IEEE/ACM International Conference on Utility and Cloud Computing (UCC), 2015.
- E. Salant, **P. Leitner**, K. Wallbom and J. Athes : *A Framework for a Cost-Efficient Cloud Ecosystem*, in eChallenges e-2015 Conference, 2015.
- M. Brandtner, **P. Leitner**, H. C. Gall: *Intent, Tests, and Release Dependencies: Pragmatic Recipes for Source Code Integration*, in Proceedings of the 15th IEEE International Working Conference on Source Code Analysis and Manipulation (SCAM), Bremen, Germany, 2015
- D. Bruneo, A. Dadashi, **P. Leitner**, B. Moltchanov, F. J. N. De-Santos, A. Miron, E. Salant, A. M. Sharifloo, K. Wallbom, and C. Woods: *CloudWave – Leveraging DevOps for Cloud Management and Application Development*, in Proceedings of the Fourth European Conference on Service-Oriented and Cloud Computing (ESOCC'15) – EU Projects Track, 2015.

- J. Cito, **P. Leitner**, T. Fritz, and H. C. Gall: *The Making of Cloud Applications – An Empirical Study on Software Development for the Cloud*, in Proceedings of the 10th Joint Meeting of the European Software Engineering Conference and the ACM SIGSOFT International Symposium on Foundations of Software Engineering (ESEC/FSE), New York, NY, USA, 2015.

- J. Cito, **P. Leitner**, H. C. Gall, A. Dadashi, A. Keller, and A. Roth: *Runtime Metric Meets Developer – Building Better Cloud Applications Using Feedback*, in Proceedings of the 2015 ACM International Symposium on New Ideas, New Paradigms, and Reflections on Programming & Software (Onward! 2015), New York, NY, USA, 2015.
- R. Zabolotnyi, **P. Leitner**, S. Schulte, and S. Dustdar: *SPEEDL – A Declarative Event-Based Language for Cloud Scaling Definition*, in Proceedings of the Future of Software Engineering For and In Cloud, Visionary Track of IEEE Services, 2015.
- G. Schermann, M. Brandtner, S. Panichella, **P. Leitner**, and H. Gall: *Discovering Loners and Phantoms in Commit and Issue Data*, in Proceedings of the 23rd IEEE International Conference on Program Comprehension (ICPC 2015), 2015.
- M. Brandtner, S. C. Müller, **P. Leitner**, and H. Gall: *SQA-Profiles: Rule-Based Activity Profiles for Continuous Integration Environments*, in Proceedings of the 22nd IEEE International Conference on Software Analysis, Evolution, and Reengineering (SANER15), 2015.
- J. Scheuner, **P. Leitner**, J. Cito, and H. Gall: *Cloud WorkBench - Infrastructure-as-Code Based Cloud Benchmarking*, in Proceedings of the 6th IEEE International Conference on Cloud Computing Technology and Science (CloudCom14), 2014.
- J. Cito, D. Suljoti, **P. Leitner**, and S. Dustdar: *Identifying Root Causes of Web Performance Degradation using Changepoint Analysis*, in Proceedings of the 14th International Conference on Web Engineering (ICWE 2014), 2014.
- R. Zabolotnyi, **P. Leitner**, and S. Dustdar: *Profiling-Based Task Scheduling for Factory-Worker Applications in Infrastructure-as-a-Service Clouds*, in Proceedings of the 40th Euromicro Conference on Software Engineering and Advanced Applications (SEAA), 2014.
- A. H. Borhani, **P. Leitner**, B. Lee, X. Li, and T. Hung: *WPress: Benchmarking Infrastructure-as-a-Service Cloud Computing Systems for On-line Transaction Processing Applications*, in Proceedings of the 18th IEEE International Enterprise Distributed Object Computing Conference (EDOC), 2014.
- C. Inzinger, W. Hummer, I. Lytra, **P. Leitner**, H. Tran, U. Zdun and S. Dustdar: *Decisions, Models, and Monitoring - A Lifecycle Model for the Evolution of Service-Based Systems*, in Proceedings of the 17th IEEE International Enterprise Computing Conference (EDOC'13), 2013
- C. Inzinger, B. Satzger, **P. Leitner**, W. Hummer, and S. Dustdar: *Position Paper: Model-based Adaptation of Cloud Computing Applications*, in: Proceedings of the International Conference on Model-Driven Engineering and Software Development (MODELSWARD), Special Track on Model-driven Software Adaptation (MODA), 2013
- C. Inzinger, W. Hummer, B. Satzger, **P. Leitner**, and S. Dustdar: *Identifying Incompatible Service Implementations using Pooled Decision Trees*, in: Proceedings of the 28th Symposium on Applied Computing (SAC), Track on Dependable and Adaptive Distributed Systems, 2013
- **P. Leitner**, C. Inzinger, W. Hummer, B. Satzger, and S. Dustdar: *Application-Level Performance Monitoring of Cloud Services Based on the Complex Event Processing Paradigm*, in: Proceedings of the 2012 IEEE International Conference on Service-Oriented Computing and Applications (SOCA'12), 2012

- W. Hummer, B. Satzger, **P. Leitner**, C. Inzinger and S. Dustdar: *Deriving a Unified Fault Taxonomy for Distributed Event-Based Systems*, in: Proceedings of the 6th ACM International Conference on Distributed Event-Based Systems (DEBS), 2012
- **P. Leitner**, W. Hummer, B. Satzger, C. Inzinger and S. Dustdar: *Cost-Efficient and Application SLA-Aware Client Side Request Scheduling in an Infrastructure-as-a-Service Cloud*, in: Proceedings of the IEEE International Conference on Cloud Computing (CLOUD'12), 2012
- **P. Leitner**, B. Satzger, C. Inzinger, W. Hummer and S. Dustdar: *CloudScale – a Novel Middleware for Building Transparently Scaling Cloud Applications*, in: Proceedings of the 27th Symposium On Applied Computing (SAC), Track on Cloud Computing, 2012
- C. Inzinger, B. Satzger, W. Hummer, **P. Leitner** and S. Dustdar: *Non-Intrusive Policy Optimization for Dependable and Adaptive Service-Oriented Systems*, in: Proceedings of the 27th Symposium On Applied Computing (SAC), Track on Dependable and Adaptive Distributed Systems, 2012
- **P. Leitner**, W. Hummer, B. Satzger and S. Dustdar: *Stepwise and Asynchronous Runtime Optimization of Web Service Compositions*, in: Proceedings of the 12th International Conference on Web Information System Engineering (WISE'11), Short Paper Track, 2011
- W. Hummer, B. Satzger, **P. Leitner** and S. Dustdar: *Distributed Continuous Queries Over Web Service Event Streams*, in: Proceedings of the 7th International Conference on Next Generation Web Services Practices (NWeSP'11), 2011
- W. Hummer, **P. Leitner**, B. Satzger and S. Dustdar: *Dynamic Migration of Processing Elements for Optimized Query Execution in Event-based Systems*, in: Proceedings of the International Symposium on Distributed Objects, Middleware, and Applications (DOA 2011), 2011
- B. Satzger, W. Hummer, **P. Leitner** and S. Dustdar: *Esc: Towards an Elastic Stream Computing Platform for the Cloud*, in: Proceedings of the 4th IEEE International Conference on Cloud Computing (CLOUD'11), Industry and Applications Track, 2011
- W. Hummer, O. Raz, O. Shehory, **P. Leitner** and S. Dustdar: *Test Coverage of Data-Centric Dynamic Compositions in Service-Based Systems*, in: Proceedings of the 4th International Conference on Software Testing, Verification and Validation (ICST), 2011
- W. Hummer, **P. Leitner** and S. Dustdar: *WS-Aggregation: Distributed Aggregation of Web Services Data*, in: Proceedings of the 26th Symposium On Applied Computing (SAC), Track on Service-Oriented Architectures and Programming , 2011
- **P. Leitner**, B. Wetzstein, D. Karastoyanova, W. Hummer, S. Dustdar and F. Leymann: *Preventing SLA Violations in Service Compositions Using Aspect-Based Fragment Substitution*, in: Proceedings of the International Conference on Service-Oriented Computing (ICSOC) , 2010
- **P. Leitner**, A. Michlmayr, F. Rosenberg and S. Dustdar: *Monitoring, Prediction and Prevention of SLA Violations in Composite Services*, in: Proceedings of the IEEE International Conference on Web Services (ICWS), Industry and Applications Track, 2010

- F. Rosenberg, M. B. Müller, **P. Leitner**, A. Michlmayr, A. Bouguettaya and S. Dustdar: *Metaheuristic Optimization of Large-Scale QoS-Aware Service Compositions*, in: Proceedings of the IEEE International Services Computing Conference (SCC), 2010
- **P. Leitner**, A. Michlmayr, F. Rosenberg and S. Dustdar: *Selecting Web Services Based on Past User Experiences*, in: Proceedings of the 2009 Asia-Pacific Services Computing Conference (APSCC 2009), 2009
- B. Wetzstein, **P. Leitner**, F. Rosenberg, I. Brandic, F. Leymann and S. Dustdar: *Monitoring and Analyzing Influential Factors of Business Process Performance*, in: Proceedings of the IEEE International Enterprise Computing Conference (EDOC), 2009
- F. Rosenberg, P. Celikovic, A. Michlmayr, **P. Leitner** and S. Dustdar: *An End-to-End Approach for QoS-Aware Service Composition*, in: Proceedings of the IEEE International Enterprise Computing Conference (EDOC), 2009
- **P. Leitner**, A. Michlmayr, F. Rosenberg, S. Dustdar: *End-to-End Versioning Support for Web Services*, in: Proceedings of the IEEE Services Computing Conference (SCC), 2008
- A. Michlmayr, F. Rosenberg, **P. Leitner** and S. Dustdar: *QoS-Aware Service Provenance in Web Service Runtimes*, in: Proceedings of the IEEE International Conference on Web Services (ICWS), 2009
- A. Michlmayr, F. Rosenberg, **P. Leitner** and S. Dustdar: *Advanced Event Processing and Notifications in Service Runtime Environments*, in: Proceedings of the IEEE International Conference on Distributed Event-Based Systems (DEBS), 2008
- R. Marin, J. Vivero, **P. Leitner**, A. Neppach and M. Zach: *Securing the Madeira Network Management System*, in: Proceedings of the IEEE International Conference on Software, Telecommunications and Computer Networks (SoftCom), 2007
- M. Leitner, **P. Leitner**, M. Zach, S. Collins and C. Fahy: *Fault Management Based on Peer-to-Peer Paradigms; A Case Study Report from the CELTIC Project Madeira*, in: Proceedings of the IFIP/IEEE International Symposium on Integrated Management (IM), 2007
- R. Marin, J. Vivero, H. Nguyen, J. Serrat, **P. Leitner**, M. Zach and C. Fahy: *A Distributed Policy Based Solution in a Fault Management Scenario*, in: Proceedings of the IEEE Global Communications Conference (GLOBECOM), Short Paper Track, 2006

WORKSHOP, POSTER, SHORT PAPER, AND DEMO CONTRIBUTIONS [\[scholar\]](#)

- J. Scheuner, and **Philipp Leitner**: *Transpiling Applications into Optimized Serverless Orchestration*, in Proceedings of the 4th International Workshops on Foundations and Applications of Self* Systems (FAS*W), 2019.
- J. Scheuner and **P. Leitner**: *Tutorial: Performance Benchmarking of Infrastructure-as-a-Service (IaaS) Clouds with Cloud WorkBench*, in Proceedings of the 16th IEEE International Conference on Autonomic Computing (ICAC), 2019.

- L. Erlenhov, F. Gomes, R. Scandariato, and **P. Leitner**: *Current and Future Bots in Software Development*, in Proceedings of the 1st International Workshop on Bots in Software Engineering (BotSE), co-located with ICSE'19. 2019.
- J. Scheuner and **P. Leitner**: *Tutorial: Performance Benchmarking of Infrastructure-as-a-Service (IaaS) Clouds with Cloud WorkBench*, in Companion Proceedings of the 10th ACM/SPEC International Conference on Performance Engineering (ICPE), 2019.
- J. Cito, **P. Leitner**, C. Bosshard, M. Knecht, G. Mazlami, and H. Gall: *PerformanceHat - Augmenting Source Code with Runtime Performance Traces in the IDE*, in Proceedings of the 40th International Conference on Software Engineering (ICSE) - Companion Volume (Demo Track), 2018.
- J. Scheuner and **P. Leitner**: *A Cloud Benchmark Suite Combining Micro and Applications Benchmarks*, in Proceedings of the 4th International Workshop on Quality-Aware DevOps (QUDOS 2018), 2018.
- C. Laaber and **P. Leitner**: *(h | g)opper: Performance History Mining and Analysis*, in Proceedings of the 7th ACM/SPEC International Conference on Performance Engineering (ICPE) – Poster and Demonstration Track, 2017.
- G. Schermann, J. Cito, **P. Leitner**, and H. C. Gall: *Towards Quality Gates in Continuous Delivery and Deployment*, in Proceedings of the 24TH IEEE International Conference on Program Comprehension (ICPC) – Short Paper Track, 2016. **Best Short Paper Award**
- J. Cito, G. Mazlami, and **P. Leitner**: *TemPerf: Temporal Correlation between Performance Metrics and Source Code*, in Proceedings of the 2nd International Workshop on Quality-Aware DevOps (QUDOS 2016), 2016.
- G. Schermann, J. Cito, and **P. Leitner**: *All the Services Large and Micro: Revisiting Industrial Practice in Services Computing*, in Proceedings of the 11th International Workshop on Engineering Service Oriented Applications (WESOA'15), co-located with ICSOC 2015, 2015.
- M. Brandtner, **P. Leitner**, and H. Gall: *Profile-based View Composition in Development Dashboards*, in Proceedings of the 17th Workshop Software-Reengineering & Evolution (WSRE), 2015.
- J. Scheuner, J. Cito, **P. Leitner**, and H. Gall: *Cloud WorkBench: Benchmarking IaaS Providers based on Infrastructure-as-Code*, in Proceedings of the 24th International World Wide Web Conference (WWW'15) - Demo Track, 2015.
- D. Bruneo, T. Fritz, S. Keidar-Barner, **P. Leitner**, F. Longo, C. Marquezan, A. Metzger, K. Pohl, A. Puliafito, D. Raz, A. Roth, E. Salant, I. Segall, M. Villari, Y. Wolfsthal, and C. Woods: *CloudWave: where Adaptive Cloud Management Meets DevOps*, in Proceedings of the Fourth International Workshop on Management of Cloud Systems (MoCS 2014), 2014.
- **P. Leitner**, S. Schulte, S. Dustdar, I. Pill, M. Schulz and F. Wotawa: *The Dark Side of SOA Testing - Towards Testing Contemporary SOAs Based on Criticality Metrics*, in: Proceedings of the 5th International Workshop on Principles of Engineering Service-Oriented Systems (PESOS), co-located with ICSE 2013, 2013

- R. Zabolotnyi, **P. Leitner** and S. Dustdar: *Dynamic Program Code Distribution in Infrastructure-as-a-Service Clouds*, in: Proceedings of the 5th International Workshop on Principles of Engineering Service-Oriented Systems (PESOS), co-located with ICSE 2013, 2013
- F. Wotawa, M. Schulz, I. Pill, S. Jehan, **P. Leitner**, W. Hummer, S. Schulte, P. Hoenisch and S. Dustdar: *Fifty Shades of Grey in SOA Testing (Position Paper)*, in: Proceedings of the 9th Workshop on Advances in Model Based Testing (A-MOST), co-located with ICST 2013, 2013
- C. Inzinger, W. Hummer, B. Satzger, **P. Leitner**, and S. Dustdar: *Towards Identifying Root Causes of Faults in Service-Based Applications*, in: Proceedings of the 31st IEEE International Symposium on Reliable Distributed Systems (SRDS 2012), Poster Track, 2012
- **P. Leitner**, W. Hummer and S. Dustdar: *A Monitoring Data Set for Evaluating QoS-Aware Service-Based Systems*, in: Proceedings of the 4th International Workshop on Principles of Engineering Service-Oriented Systems (PESOS), co-located with ICSE'12, 2012
- **P. Leitner**, F. M. Nardini, G. Tolomei, F. Silvestri and S. Dustdar: *Mining Lifecycle Event Logs for Enhancing Service-based Applications*, in: Proceedings of the 1st International Workshop on Adaptive Services for the Future Internet (WAS4FI), colocated with ServiceWave'11, 2011
- W. Hummer, **P. Leitner** and S. Dustdar: *A Step-By-Step Debugging Technique To Facilitate Mashup Development and Maintenance*, in: Proceedings of the 4th International Workshop on Web APIs and Services Mashups (Mashups'10) co-located with ECOWS 2010, 2010
- **P. Leitner**, B. Wetzstein, F. Rosenberg, A. Michlmayr, S. Dustdar and F. Leymann: *Runtime Prediction of Service Level Agreement Violations for Composite Services*, in: Proceedings of the 3rd Workshop on Non-Functional Properties and SLA Management in Service-Oriented Computing (NFPSLAM'09), co-located with ICSOC 2009, 2009
- A. Michlmayr, F. Rosenberg, **P. Leitner** and S. Dustdar: *Comprehensive QoS Monitoring of Web Services and Event-Based SLA Violation Detection*, in: Proceedings of the 4th International Workshop on Middleware for Service Oriented Computing (MW4SOC), co-located with MIDDLEWARE'09, 2009
- I. Brandic, D. Music, **P. Leitner**, S. Dustdar: *VieSLAF Framework: Enabling Adaptive and Versatile SLA-Management*, in: Proceedings of the 6th International Workshop on Grid Economics and Business Models (Gencon), co-located with Euro-Par'09, 2009
- F. Rosenberg, **P. Leitner**, A. Michlmayr, P. Celikovic and S. Dustdar: *Towards Composition as a Service - A Quality of Service Driven Approach*, in: Proceedings of the 1st Workshop on Information and Software as Service (WISS), co-located with ICDE'09, 2009
- **P. Leitner**, A. Michlmayr and S. Dustdar: *Towards Flexible Interface Mediation for Dynamic Service Invocations*, in: Proceedings of the 3rd Workshop on Emerging Web Services Technology (WEWST), co-located with ECOWS'08, 2008
- F. Rosenberg, **P. Leitner**, A. Michlmayr and S. Dustdar: *Integrated Metadata Support for Web Service Runtimes*, in: Proceedings of the Middleware for Web Services Workshop (MWS), co-located with EDOC'08, 2008

EDITED BOOKS [\[scholar\]](#)

- A. Celesti and **P. Leitner** (Eds.): *Advances in Service-Oriented and Cloud Computing – Workshops of ESOC 2015, Revised Selected Papers*. Communications in Computer and Information Science (CCIS 567). ISBN: 3319333127. 2016.

CONTRIBUTIONS TO BOOKS [\[scholar\]](#)

- Rob van der Mei, Hans van den Berg, Ivan Ganchev, Kurt Tutschku, **Philipp Leitner**, Pasi Lassila, Wojciech Burakowski, Fidel Liberal, Ake Arvidsson, Tobias Hossfeld, Katarzyna Wac, Hugh Melvin, Tihana Galinac Grbac, Yoram Haddad, and Peter Key: *State of the Art and Research Challenges in the Area of Autonomous Control for a Reliable Internet of Services*, in: *Autonomous Control for a Reliable Internet of Services*, Springer, 2018.
- W. Hummer, **P. Leitner**, A. Michlmayr, F. Rosenberg, S. Dustdar: *VRESCo – Vienna Runtime Environment for Service-oriented Computing*, in: *Service Engineering*. European Research Results, Springer, 2010
- A. Michlmayr, **P. Leitner**, F. Rosenberg and S. Dustdar: *Event Processing in Web Service Runtime Environments*, in: *Handbook of Research on Advanced Distributed Event-Based Systems*, Publish/Subscribe and Message Filtering Technologies, IGI Global, 2010
- **P. Leitner**, F. Rosenberg, A. Michlmayr, A. Huber and S. Dustdar: *A Mediator-Based Approach to Resolving Interface Heterogeneity of Web Services*, in: *Emerging Web Services Technology*, Volume 3, Birkhäuser, 2009
- C. Fahy, M. Ponce de Leon, S. van der Meer, R. Marin, J. Vivero, J. Serrat, N. Georgalas, **P. Leitner**, S. Collins and B. Baesjou: *Modelling Behaviour and Distribution for the Management of Next Generation Networks*, in *Whitestein Series in Software Agent Technologies and Autonomic Computing*, Special issue on Advanced Autonomic Networking and Communication, Birkhäuser, 2008

INVITED PUBLICATIONS, EDITORIALS, AND OTHER NON-REVIEWED MATERIAL [\[scholar\]](#)

- F. Markusse, **P. Leitner**, and Alexander Serebrenik: *Towards Continuous Performance Assessment of Java Applications With PerfBot*, Proceedings of the 4th International Workshop on Bots in Software Development (BotSE), invited contribution, 2023
- U. Zdun, E. Wittern, and **P. Leitner**: *Emerging Trends, Challenges, and Experiences in DevOps and Microservice APIs*, IEEE Software 37(1): 87-91 (2020)
- A. van Hoorn, P. Jamshidi, **P. Leitner**, and I. Weber: *Report from GI-Dagstuhl Seminar 16394: Software Performance Engineering in the DevOps World*, arXiv:1709.08951 [cs.PF], 2018
- E. Elmroth, **P. Leitner**, S. Schulte, S. Venugopal: *Connecting Fog and Cloud Computing*, IEEE Cloud Computing, vol. 4, nr. 2, pp. 22 - 25, 2017.

- R. Ranjan, R. Buyya, **P. Leitner**, A. Haller, and S. Tai: *A Note on Software Tools and Techniques for Monitoring and Prediction of Cloud Services*, Software: Practice and Experience, 2014.
- B. Satzger, W. Hummer, C. Inzinger, **P. Leitner**, and S. Dustdar, *Winds of Change: From Vendor Lock-In to the Meta Cloud*, IEEE Internet Computing, vol. 17, no. 1, pp. 69-73, Jan.-Feb., 2013
- S. Tai, **P. Leitner**, S. Dustdar, *Design by Units: Abstractions for Human and Compute Resources for Elastic Systems*, IEEE Internet Computing, vol. 16, no. 4, pp. 84-88, July-Aug. 2012, doi:10.1109/MIC.2012.81
- **P. Leitner**, S. Dustdar, B. Wetzstein, and F. Leymann: *Cost-Based Prevention of Violations of Service Level Agreements in Composed Services Using Self-Adaptation*, in: Proceedings of the S-Cube@ICSE Workshop, co-located with ICSE'12, 2012

THESES [\[scholar\]](#)

- **P. Leitner**: *On Preventing Violations of Service Level Agreements in Composed Services using Self-Adaptation*, PhD Thesis, Vienna University of Technology. Primary advisor: Prof. Schahram Dustdar. Examiner: Prof. Fabio Casati. 2011.
- **P. Leitner**: *The Daios Framework – Dynamic, Asynchronous and Message-oriented Invocation of Web Services*, Master's Thesis, Vienna University of Technology. Primary advisor: Prof. Schahram Dustdar. Responsible assistant: Florian Rosenberg. 2007.